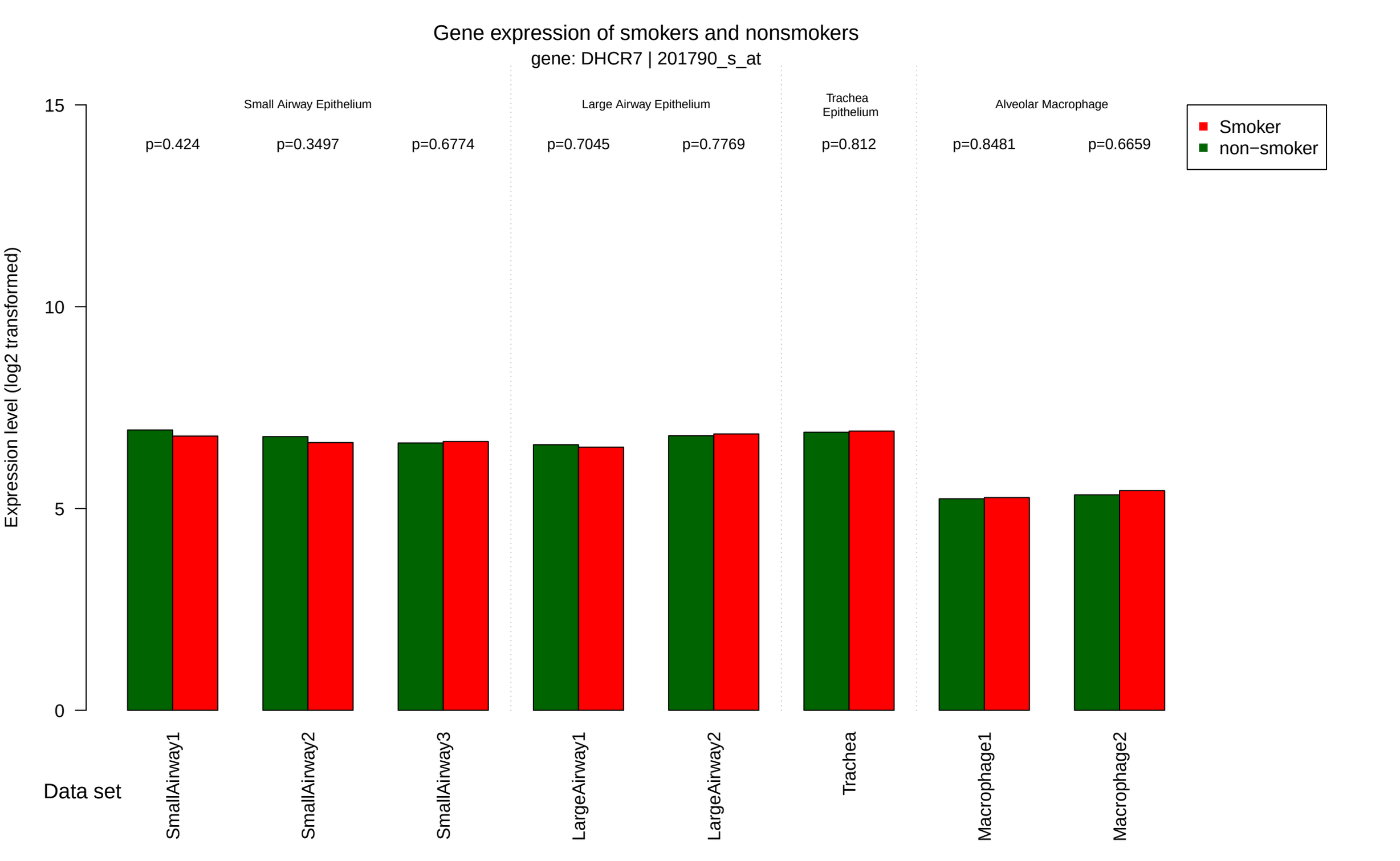
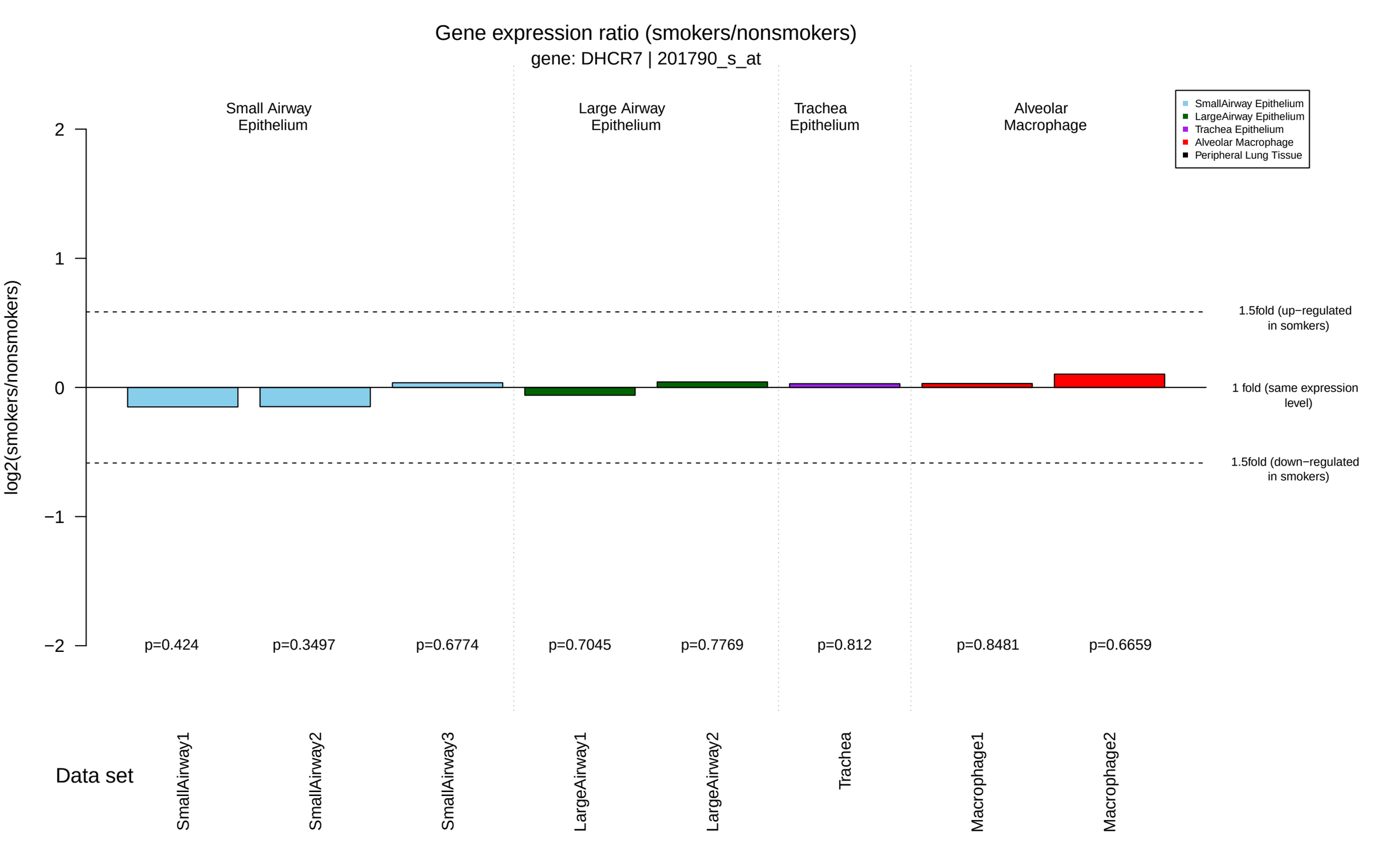
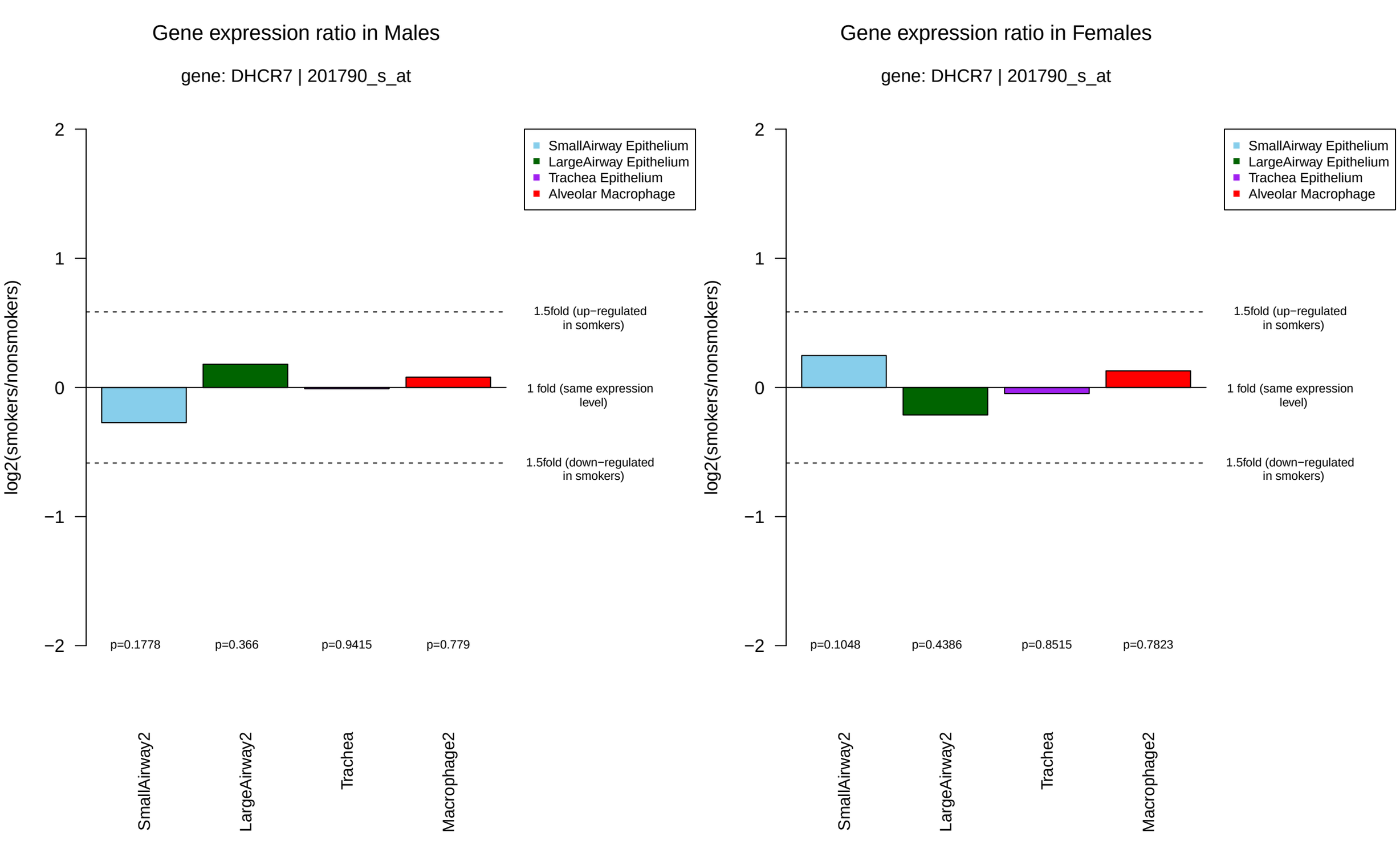
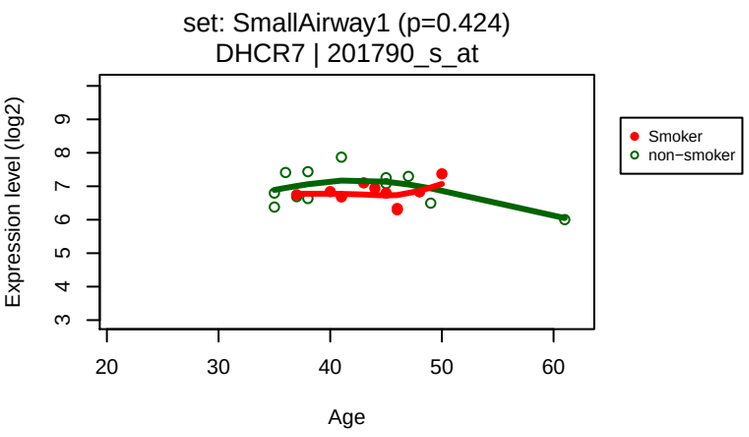
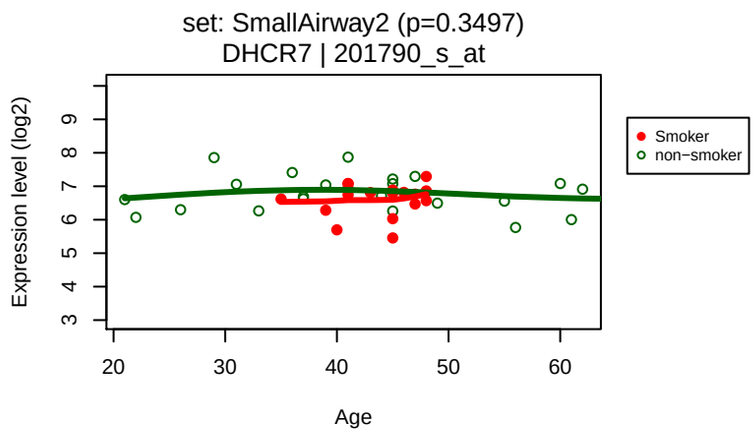
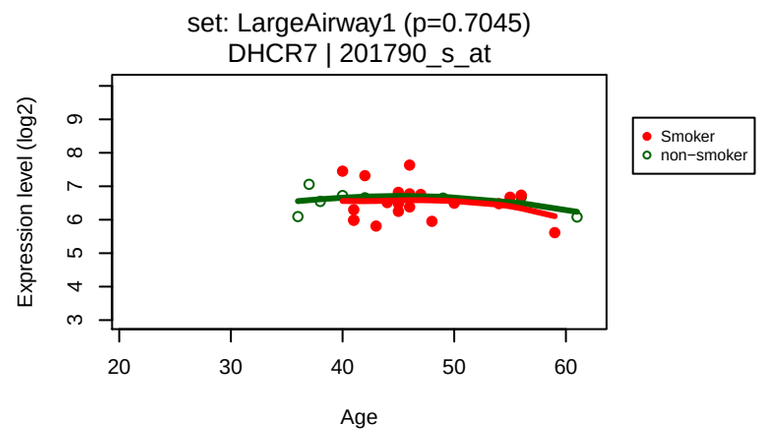

Gene expression of smokers and nonsmokers
gene: DHCR7 | 201790_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.424
p=0.3497
p=0.6774
p=0.7045
p=0.7769
p=0.812
p=0.8481
p=0.6659
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

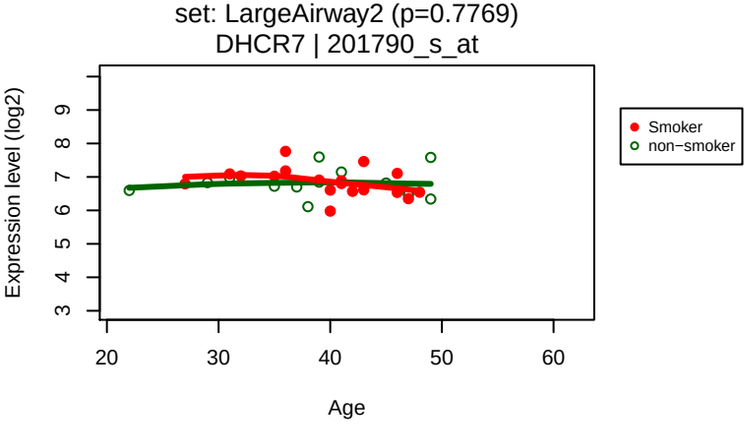

Gene expression ratio (smokers/nonsmokers)
gene: DHCR7 | 201790_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.424
p=0.3497
p=0.6774
p=0.7045
p=0.7769
p=0.812
p=0.8481
p=0.6659
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

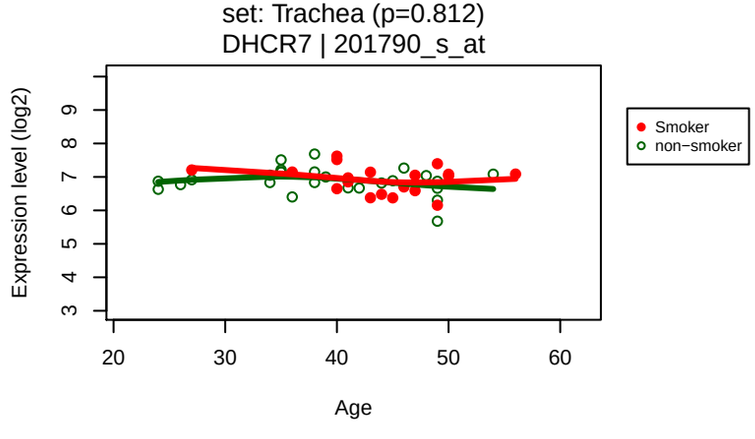

Gene expression ratio in Males
gene: DHCR7 | 201790_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1778
p=0.366
p=0.9415
p=0.779
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DHCR7 | 201790_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1048
p=0.4386
p=0.8515
p=0.7823
SmallAirway2
LargeAirway2
Trachea
Macrophage2

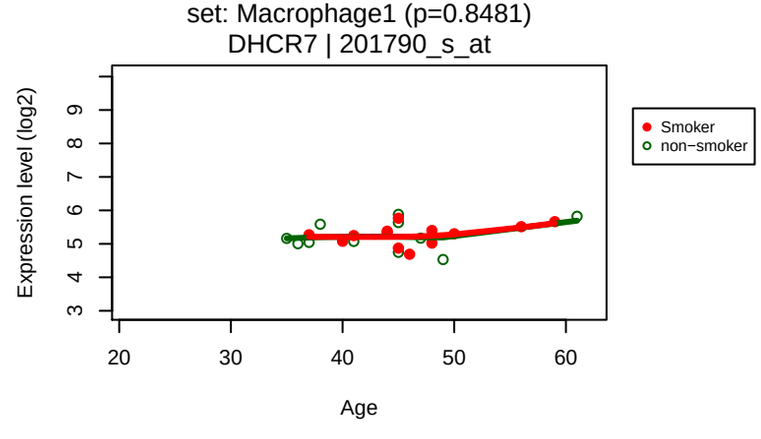

set: SmallAirway1 (p=0.424)
DHCR7 | 201790_s_at
Expression level (log2)
Age
Smoker
non−smoker

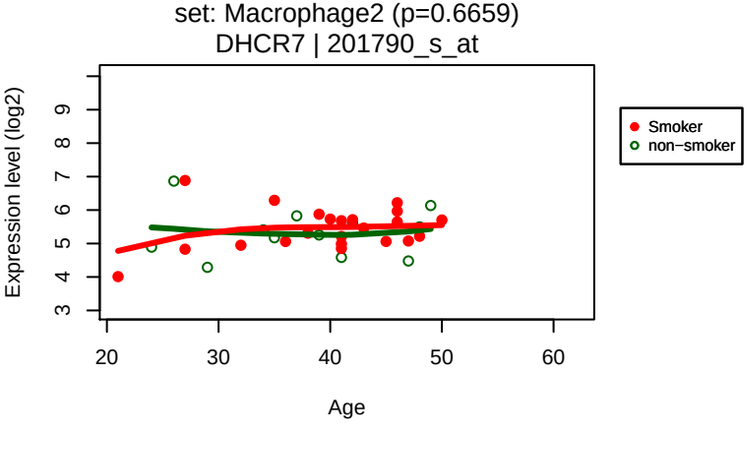

set: SmallAirway2 (p=0.3497)
DHCR7 | 201790_s_at
Expression level (log2)
Age
Smoker
non−smoker

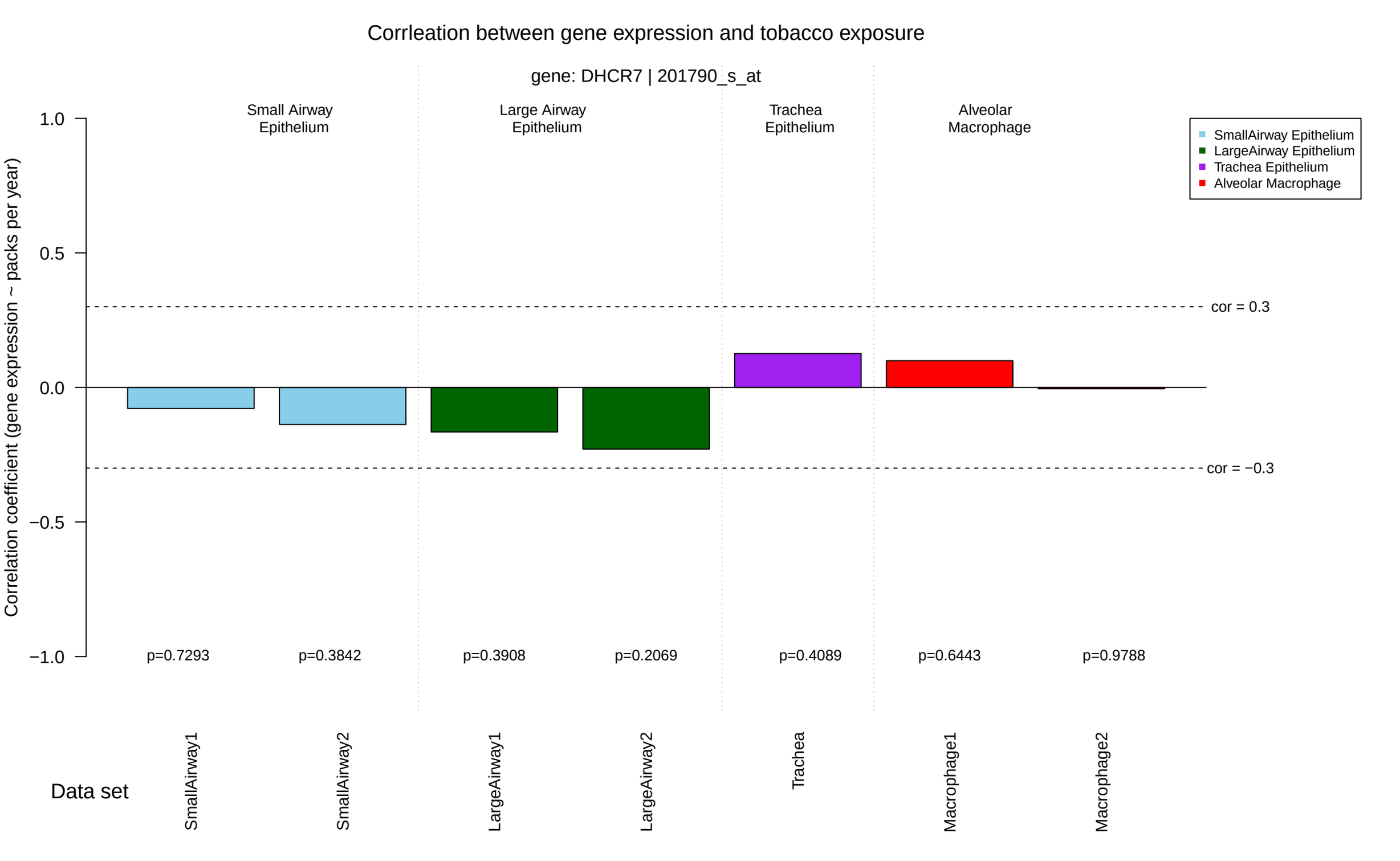

set: LargeAirway1 (p=0.7045)
DHCR7 | 201790_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7769)
DHCR7 | 201790_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.812)
DHCR7 | 201790_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8481)
DHCR7 | 201790_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6659)
DHCR7 | 201790_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DHCR7 | 201790_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7293
p=0.3842
p=0.3908
p=0.2069
p=0.4089
p=0.6443
p=0.9788
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2